

Assignment

Statistics

→ Introduction

→ Advantages / usages

→ ~~us~~ where used / applied.Measurement of central ~~contrasts~~ tendency.

- primary data - unorganized data.
Ex: classmates' age.
- secondary data - It help to take a decision, useful take an information.
Ex: financial information.
- Numerical data - number based. / using
Ex: Age. in calculation.
- Un-numerical data - character based.
Ex: Breakfast.
- Data without frequency
23, 24, 20, 32, 35, 26, 26, 28
un organized
- Data with frequency - organized.

x	⇒	20	23	24	26	26	28	32	35
f	⇒	1	1	1	2	1	1	1	1

 = 8
- Group data.
class interver.

How many times a data occurs - frequency.

calculation of mean / average

Data without frequency

$$\bar{x} = \frac{\sum x}{n}$$

n - number of data

$\sum x$ - Total of x

$\bar{x}$ - mean.

01. 25, 30, 36, 40, 28, 18, 20,
22, 18, 21 calculate mean.

$$\bar{x} = \frac{\sum x}{n}$$

$$= \frac{256}{10}$$

$$= 25.6$$

02. The details of science marks obtain by 20 students in a task are given below.

45, 62, 35, 70, 29, 50, 62, 90,
57, 45, 54, 5
62, 70, 75, 32, 38, 59, 45, 16,
39, 50

Calculate mean.

$$\bar{X} = \frac{\sum x}{n}$$

$$= \frac{1031}{20}$$

$$= 51.55$$

Data with frequency

$$\bar{X} = \frac{\sum fx}{\sum f}$$

<u>x</u>	<u>f</u>	<u>fx</u>	$\sum fx = 765$
10	4	40	$\sum f = 32$
15	6	90	
20	8	180	= 765
25	2	50	
30	4	120	= 23.90625
35	3	105	
40	5	200	

Calculate mean $32 = \sum f$ 765

Grouped Data

Ex 1: Class Interval frequency (f)

5 - 9

4

10 - 14

5

15 - 19

9

⊙ limits $\Rightarrow$

• lower limit - 10

• Upper limit - 14

⊙ boundary $\Rightarrow$

• lower boundary - $\frac{9 + 10}{2} = 9.5$

• upper boundary - $\frac{14 + 15}{2} = 14.5$

⊙ class size ~~weight~~ ^{width} $\Rightarrow$

= upper boundary - lower boundary

= 14.5 - 9.5

= 5

OR

= (upper limit - lower limit) + 1

$$= (14 - 10) + 1$$

$$= \underline{\underline{5}}$$

① midpoint $\Rightarrow$

$$= \frac{\text{lower limit} + \text{upper limit}}{2}$$

$$= \frac{10 + 14}{2} = \underline{\underline{12}}$$

OR

$$= \frac{\text{lower boundary} + \text{upper boundary}}{2}$$

$$= \frac{9.5 + 14.5}{2}$$

$$= \underline{\underline{12}}$$

Ex 2: 20, 25, 56, 49, 62, 38, 88, 22,
10, 27, ✓ ✓

45, 46, 52, 21, 55, 57, 54, 12,
38, 41 ✓ ✓

32, 22, 17, 25, 35, 38, 41, 60,
16, 22 ✓ ✓

60, 58, 09, 11, 26, 22, 35, 37,
41, 48 ✓ ✓

Prepare frequency distribution
table using the (first class interval
starting from 9 - 16).

class interval frequency

9	-	16	 1	6
17	-	24	 	8
25	-	32	 1	5
33	-	40	 1	6
41	-	48	 1	6
49	-	56		4
57	-	64	 	5

$$\Sigma f = 40$$

Ex3: Prepare frequency distribution table using (class size = 5).

class interval frequency

9	- 13		4
14	- 18		4
19	- 23	 	6
24	- 28		4
29	- 33		1
34	- 38	 	6
39	- 43		3
44	- 48		3
49	- 53	 	2
54	- 58		4
59	- 63		3
			40

Calculation of mean.

$$\bar{X} = \frac{\sum fx}{\sum f}$$

$u \Rightarrow$ midpoint.

$$\left[\frac{\text{lower limit} + \text{upper limit}}{2} \right]$$

$$\left[\frac{9 + 13}{2} \right]$$

X	f	fu
11	4	44
16	4	64
21	6	126
26	4	104
31	1	31
36	6	216
41	3	123
46	3	138
51	2	102
56	4	224
61	3	183
	<u>40</u>	<u>1355</u>

$$\bar{X} = \frac{\sum fx}{\sum f}$$

$$= \frac{1355}{40}$$

$$= \underline{\underline{33.875}} / \underline{\underline{33.88}}$$

Ex 4: 26, 32, 35, 39, 18, 16, 25,
10, 25, 32 ✓

(63), 59, 41, 46, 44, 39, 40,
55, 57, 28 ✓

17, 22, 28, 26, 19, 55, 54,
60, 62, 58 ✓

42, 48, 50, 40, 49, 36, 28,
15, 52, 58

10, 17, 16, 21, 29, 31, 34,
43, 51, 62

prepare frequency distribution table for the above data using (first class interval 10-14). Calculate mean.

Class Interval		frequency
10 - 14	II	2
15 - 19	IIII II	7
20 - 24	II	2
25 - 29	IIII III	8
30 - 34	IIII	4
35 - 39	IIII	4
40 - 44	IIII I	6
45 - 49	III	3
50 - 54	IIII	4
55 - 59	IIII I	6
60 - 64	IIII	4
		50

$$\textcircled{1} \quad X = \frac{10 + 14}{2} = 12.$$

} 5

$$\textcircled{2} \quad X = \frac{15 + 19}{2} = 17.$$

} 5

$$\textcircled{3} \quad X = \frac{20 + 24}{2} = 22$$

X	f	fX
12	2	24
17	7	119
22	2	44
27	8	216
32	4	128
37	4	148
42	6	252
47	3	141
52	4	208
57	6	342
62	4	248
67		
	<u>50</u>	<u>1870</u>

$$\bar{X} = \frac{\sum fX}{\sum f}$$

$$= \frac{1870}{50} = 37.4$$

Calculation of median.

* mid value of arranging data.

Data without frequency

Ex1. 22, 26, 08, 09, 31, 36, 42
findout the median.

08, 09, 22, 26, 31, 36, 42.

↑
median.

median = 26

Ex2: 46, 48, 50, 56, 58, 60, 61, 70,
78, 80, 82, 84 findout the median

$$\begin{aligned} \text{median} &= \frac{60 + 61}{2} = \frac{(n+1)^{\text{th}}}{2} \\ &= 60.5 \end{aligned}$$

Data with frequency

Ex1.

<u>X</u>	<u>f</u>
5	6
10	4
15	3
20	2
25	5
30	8
35	7
40	2

findout the median.

Cumulative frequency.

<u>X</u>	<u>f</u>	<u>Cf</u>
5	6	6
10	4	10
15	3	13
20	2	15
25	5	20
30	8	28
35	7	35
40	2	37

$$\frac{\sum f}{2} = \frac{37}{2}$$

<<include>>

$$= 18.5$$

$$\text{Median} = \underline{\underline{25}}$$

Ex 2:

<u>X</u>	<u>f</u>
3	5
6	6
9	4
12	8
15	3
18	9
21	5
24	6
27	2
30	7

findout the median .

<u>X</u>	<u>f</u>	<u>cf</u>
3	5	5
6	6	11
9	4	15
12	8	23
15	3	26
18	9	35
21	5	40
24	6	46
27	2	48
30	7	55

$$\begin{aligned} \Sigma f &= \frac{55}{2} \\ &= 27.5 \end{aligned}$$

Median = 18

Ex 3:

<u>X</u>	<u>f</u>
6✓	4
3✓	3
5✓	4
2✓	6
8✓	2

findout the median.

<u>X</u>	<u>f</u>	<u>Cf</u>
2	6	6
3	3	9
5	4	13
6	4	17
8	2	19

$$\Sigma f = \frac{19}{2}$$

$$= 9.5$$

$$\text{Median} = \underline{\underline{5}}$$

Grouped Data

$$\text{Median} = L + \frac{C}{f} \left(\frac{\Sigma f}{2} - PC \right)$$

$L \Rightarrow$ Lower boundary of median class

$C \Rightarrow$ Class size / width.

$f \Rightarrow$ frequency of median class.

$PC \Rightarrow$ Cumulative frequency of previous class interval.

Ex 1.

<u>Class Interval</u>	<u>f</u>
5 - 9	6
10 - 14	8
15 - 19	5
20 - 24	7
25 - 29	8
30 - 34	3
35 - 39	9
40 - 44	4
45 - 49	2

find out the median.

Median class

<u>Class Interval</u>	<u>f</u>	<u>cf</u>
5 - 9	6	6
10 - 14	8	14
15 - 19	5	19
20 - 24	7	26
25 - 29	8	34
30 - 34	3	37
35 - 39	9	46
40 - 44	4	50
45 - 49	2	52

$$\frac{\sum f}{2} = \frac{52}{2} = 26$$

$$L = \frac{19 + 20}{2} = 19.5$$

$$C = (9 - 5) + 1 = 5$$

$$f = 7$$

$$PC = 19$$

$$\text{Median} = L + C/f \left(\frac{\sum f}{2} - PC \right)$$

$$= 19.5 + \frac{5}{7} (26 - 19)$$

$$= 19.5 + \left(\frac{5}{7} \times 7 \right)$$

$$= 19.5 + \frac{35}{7}$$

$$= 24.5$$

Ex2: class interval f

8 - 13	4
14 - 19	6
20 - 25	3
26 - 31	5
32 - 37	2
38 - 43	7
44 - 49	8
50 - 55	4
56 - 61	

findout the median.

<u>Class Interval</u>	<u>f</u>	<u>Cf</u>
8 - 13	4	4
14 - 19	6	10
20 - 25	3	13
26 - 31	5	18
32 - 37	2	20
38 - 43	7	27
44 - 49	8	35
50 - 55	4	39

$$\frac{\sum f}{2}$$

$$= \frac{39}{2}$$

$$= 19.5$$

$$C = 6$$

$$f = 2$$

$$L = 31.5$$

$$PC = 18$$

$$\text{Median} = L + \frac{C}{f} \left(\frac{\sum f}{2} - PC \right)$$

$$= 31.5 + \frac{6}{2} (19.5 - 18)$$

$$= 31.5 + 3 (1.5)$$

$$= 31.5 + 4.5$$

$$= 36$$

Calculation of mode

data which time more occurs.

Data without frequency

Ex1: 5, 6, 8, 6, 7, 2, 4, 9.

Mode = 6.

Ex2: 5, 3, 7, 8, 7, 6, 4, 4, 3, 4.

Mode = 4

Ex3: 8, 9, 6, 5, 9, 8, 3

Mode = 8 and 9

Ex4: 5, 4, 3, 7, 10, 6

Mode = There is no mode.

Data with frequency

Ex1:

X	f
5	4
7	6
9	3
11	4
13	8
15	2
17	2
21	7

Mode = 13

Grouped data

$$\text{Mode} = L + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2} \right) \times C.$$

$L \Rightarrow$ Low boundary of model class

$\Delta_1 \Rightarrow$ The difference between frequency of model class and frequency of previous class

$\Delta_2 \Rightarrow$ The difference between frequency of model class and frequency of next class.

$C \Rightarrow$ Class size / width.

Ex: Class Interval f

7 - 10	4	
11 - 14	5	} 3 = \Delta_1
15 - 18	8	
19 - 22	6	} 2 = \Delta_2
23 - 26	3	
27 - 30	4	
31 - 34	2	
35 - 38	4	

findout mode for the above data.

$$\text{Mode} = L + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2} \right) \times C$$

$$= 14.5 + \left(\frac{3}{3 + 2} \right) \times 4$$

$$= 14.5 + \left(\frac{3}{5} \times 4 \right)$$

$$= 16.9$$

(2020) - Part II

Q1. Following information represent daily income of 150 families:

Income (Rs) No. of families.

More than Rs. 1000

150

More than Rs. 1500

140

> Rs. 2000

115

> Rs. 2500

95

> Rs. 3000

65

> Rs. 3500

53

> Rs. 4000

32

> Rs. 4500

15

95
40
135
15
4

- i. Re arrange the data frequency distribution table
- ii. Calculate the mode & median.

i. Class Interval

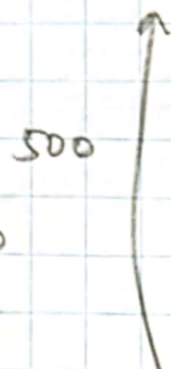
500	-	1000
1000	-	1500
1500	-	2000
2000	-	2500
2500	-	3000
3000	-	3500
3500	-	4000
4000	-	4500
4500	-	5000

$$\frac{1501 + 2000}{2}$$

$$\frac{3501}{2} = 1750.5$$

$$\frac{1001 + 1500}{2}$$

$$\frac{2501}{2} = 1250.5$$



Class Interval

fx

f

x

1001 - 1500 1250.5 10 1250.5

1501 - 2000 1750.5 25 1750.5

2001 - 2500 2250.5 20 2250.5

2501 - 3000 2750.5 30 2750.5

3001 - 3500 3250.5 12 3250.5

3501 - 4000 3750.5 11 3750.5

4001 - 4500 4250.5 7 4250.5

4501 - 5000 4750.5 15 4750.5

$\Sigma fx = 150$

$$\begin{aligned}\text{Median} &= \frac{\sum f}{\sum fx} \\ &= \underline{150}\end{aligned}$$

$$\begin{aligned}\text{Median} &= \frac{\sum fx}{\sum f} \\ &= \frac{445075}{150} \\ &= 2967.167\end{aligned}$$

$$\begin{aligned}\text{Mode} &= L + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2} \right) \times C \\ &= 2500.5 + \left(\frac{10}{10 + 18} \right) \times 500 \\ &= 2500.5 + \frac{10}{28} \times 500 \\ &= 2500.5 + 178.57 \\ &= 2679.071\end{aligned}$$

Measures of dispersion.

Data with frequency

Ex1.

<u>X</u>	<u>f</u>
2	5
4	3
6	2
8	6
10	4
12	1
14	7
16	2
18	4

Calculate median, Q_3 , D_5
 P_{90}

<u>X</u>	<u>f</u>	<u>cf</u>
2	5	5
4	3	8
6	2	10
8	6	16
10	4	20
12	1	21
14	7	28
16	2	30
18	4	34

$$\text{Median} = \frac{\sum f}{2}$$

$$= \frac{34}{2}$$

$$= 17^{\text{th}}$$

Median is 10.

$$\begin{aligned}
 Q_3 &= \frac{k \varepsilon f}{4} \\
 &= \frac{3 \times 34}{4} \\
 &= \frac{102}{4} \\
 &= 25.5 \text{ th}
 \end{aligned}$$

$$Q_3 = \underline{\underline{14}}$$

$$\begin{aligned}
 D_5 &= \frac{k \varepsilon f}{10} \\
 &= \frac{5 \times 34}{10} \\
 &= \underline{\underline{10.2 \text{ th}}}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{170}{10} \\
 &= \underline{\underline{17 \text{ th}}}
 \end{aligned}$$

$$D_5 = \underline{\underline{10}}$$

$$\begin{aligned}
 P_{90} &= \frac{k \varepsilon f}{100} \\
 &= \frac{90 \times 34}{100} \\
 &= 30.6 \text{ th}
 \end{aligned}$$

$$= 306$$

$$P_{90} = \underline{\underline{18}}$$

Standard Deviation

It calculate for data between the mean.

Data without frequency

$$SD(\sigma) = \sqrt{\frac{\sum u^2}{n} - (\bar{u})^2} \rightarrow \left(\frac{\sum x}{n} \right)^2$$

$$(SD)^2 = \text{variance}$$

$$\text{Variance} = \frac{\sum u^2}{n} - (\bar{u})^2$$

Ex1:

~~5~~
~~8~~
 10
 13
~~4~~
~~7~~
~~6~~
~~9~~
~~5~~
 11

Calculate mean, standard deviation.

x	x^2
4	16
5	25
5	25
6	36
7	49
8	64
9	81
10	100
11	121
13	169
78	$\Sigma x^2 = 686$

$$\begin{array}{r} 13 \\ 13 \\ \hline 39 \\ 13 \\ \hline 169 \end{array}$$

$$\text{Mean}(\bar{x}) = \frac{\Sigma x}{n}$$

$$= \frac{78}{10}$$

$$= 7.8$$

$$(SD) \sigma = \sqrt{\frac{\Sigma x^2}{n} - (\bar{x})^2}$$

$$= \sqrt{\frac{686}{10} - (7.8)^2}$$

$$= \sqrt{68.6 - 60.84}$$

$$= \sqrt{7.76}$$

$$= 2.79$$

Ex2: ~~16~~, 20, 25, 36, ~~11~~, ~~9~~, ~~7~~, ~~8~~, ~~4~~,
~~6~~, ~~13~~, ~~15~~, 22 Calculate SD & mean

<u>X</u>	<u>X²</u>
4	16
6	36
7	49
8	64
9	81
10	100
11	121
13	169
15	225
20	400
22	484
25	625
36	1296

$$\sum X^2 = 3666$$

$$\text{Mean} = \frac{\sum X}{n}$$

$$= \frac{186}{13}$$

$$= 14.31$$

$$\begin{aligned}
 SD &= \sqrt{\frac{\sum x^2}{n} - (\bar{x})^2} \\
 &= \sqrt{\frac{3666}{13} - (204.78)^2} \\
 &= \sqrt{282 - 204.78} \\
 &= \sqrt{77.22} \\
 &= 8.79
 \end{aligned}$$

Data with frequency

$$SD(\sigma) = \sqrt{\frac{\sum f u^2 x^2}{\sum f} - (\bar{u})^2}$$

Ex1.	<u>X</u>	<u>f</u>
	2	5
	4	3
	6	2
	8	6
	10	7
	12	4
	14	8
	16	10
	18	9
	20	3

Calculate mean & SD.

<u>x</u>	<u>f</u>	<u>fx</u>	<u>x²</u>	<u>fx²</u>
2	5	10	4	20
4	3	12	16	48
6	2	12	36	72
8	6	48	64	384
10	7	70	100	700
12	4	48	144	576
14	8	112	196	1568
16	10	160	256	2560
18	9	162	324	2916
20	3	60	400	1200
<u>$\Sigma f = 57$</u>		<u>694</u>	<u>400</u>	<u>12000</u>

$$\Sigma fx = 694$$

$$\Sigma fx^2 = 10044$$

$$\text{Mean} = \frac{\Sigma fx}{\Sigma f}$$

$$= \frac{694}{57}$$

$$= 12.18$$

$$6 = \sqrt{\frac{\Sigma fx^2}{\Sigma f} - (\bar{x})^2}$$

$$= \sqrt{\frac{10044}{57} - (12.18)^2}$$

$$= \sqrt{176.21 - 148.35}$$

$$= \sqrt{27.86} = 5.28$$

Grouped Data.

$$SD = \sqrt{\frac{\sum fx^2}{\sum f} - (\bar{x})^2}$$

Ex1: Class Interval f

6 - 9	4
10 - 13	7
14 - 17	3
18 - 21	8
22 - 25	5
26 - 29	2
30 - 33	1
34 - 37	6
38 - 41	4

Calculate Mean & SD.

<u>Class Interval</u>	<u>f</u>	<u>X</u>	<u>fx</u>	<u>x²</u>	
6 - 9	4	7.5	30	56.25	$\frac{6+9}{2} = 7.5$
10 - 13	7	11.5	80.5	132.25	$\frac{2fx^2}{\sum f} = \frac{15}{2} = 7.5$
14 - 17	3	15.5	46.5	240.25	
18 - 21	8	19.5	156	380.25	
22 - 25	5	23.5	117.5	552.25	
26 - 29	2	27.5	55	756.25	
30 - 33	1	31.5	31.5	992.25	
34 - 37	6	35.5	213	1260.25	
38 - 41	4	39.5	158	1560.25	
	<u>$\sum f = 40$</u>		<u>$\sum fx = 888$</u>		<u>$\sum fx^2 = 79370.875$</u>

$$\text{Mean} = \frac{\sum fx}{\sum f}$$

$$= \frac{22.2}{\cancel{888}} \\ 40$$

$$= 22.2.$$

$$\sigma = \sqrt{\frac{\sum fx^2}{\sum f} - (\bar{x})^2}$$

$$= \sqrt{\frac{79370.875}{40} - (22.2)^2}$$

$$= \sqrt{1984.27 - 492.84}$$

$$= \sqrt{1491.43}$$

$$= \underline{\underline{34.52}}$$

2017

The mean & SD of 20 items were calculated by a student as 40 & 10 respectively. But while calculating them each item 15 was misrate as 50 find the correct mean & SD.

$$\text{Mean } (\bar{x}) = \frac{\sum fx}{\sum f}$$

$$40 = \frac{\sum fx}{20}$$

$$800 = \sum fx$$

$$SD = \sqrt{\frac{\sum fx^2}{\sum f} - (\bar{x})^2}$$

$$10 = \sqrt{\frac{\sum fx^2}{20} - 1600}$$

$$100 + 1600 \times 20 = \sum fx^2$$

$$34000 = \sum fx^2$$

$$\sum fx = 800 - 50 + 15$$

$$= 765$$

$$\sum fx^2 = 34000 - 50 + 15$$

$$= 31725$$

$$\text{Mean} = \frac{765}{20}$$

$$= 38.25$$

$$SD = \sqrt{\frac{31725}{20} - 1463.0625}$$

$$= \sqrt{1586.25 - 1463.0625}$$

$$= \sqrt{123.19}$$

$$= 11.10$$

Q2. X, Y companies are software development companies and following details are number of software sold during the year 2020 from January to December.

X company			Y company		
Month	x^2	No. of software sold $\bar{x}$	Month	y^2	No. of software $\bar{y}$
Jan	40000	200	Jan	40000	200
Feb	90000	300	Feb	62500	250
Mar	22500	150	Mar	32400	180
Apr	10000	100	Apr	48400	220
May	5625	75	May	55225	235
Jun	7225	85	Jun	30625	175
Jul	8100	90	Jul	15625	125
Aug	14400	120	Aug	56644	238
Sep	32400	180	Sep	12544	112
Oct	14884	122	Oct	93025	305
Nov	13924	118	Nov	50625	225
Dec	67600	260	Dec	18225	135
				<u>515838</u>	<u>2400</u>
	<u>326658</u>	<u>1800</u>			

Calculate Coefficient of variation for both companies & explain which company ~~has~~ more variability in its sales of software programs.

X company $\Rightarrow$ Y company $\Rightarrow$

$$\bar{X} = \frac{\sum X}{n}$$

$$= \frac{1800}{12}$$

$$= 150$$

$$SD = \sqrt{\frac{\sum X^2}{n} - (\bar{X})^2}$$

$$= \sqrt{\frac{326658}{12} - 22500}$$

$$= \sqrt{27221.5 - 22500}$$

$$= \sqrt{4721.5} = 68.71$$

$$CV = \frac{SD}{\bar{X}} \times 100$$

$$= \frac{68.71}{150} \times 100$$

$$= 45.8 \quad (46)$$

Y company $\Rightarrow$

$$\bar{X} = \frac{\sum x}{n} = \frac{2400}{12}$$

$$= 200$$

$$SD = \sqrt{\frac{\sum x^2}{n} - (\bar{X})^2}$$

$$= \sqrt{\frac{515838}{12} - 40000}$$

$$= \sqrt{42986.5 - 40000}$$

$$= \sqrt{2986.5} = 54.65$$

$$CV = \frac{54.65}{200} \times 100$$

$$= 27.325$$

(28)